

SES bridge mag null policy

Core module: `src/lib/seismic/ses-bridge.ts`

Problem it solves

GOSSIP often publishes **Md = N/D**. If those become `mag: 0`, SES min-mag filters and “max M” stats treat them as real M0 energy. If they are **dropped**, densify under swarm load looks empty. The bridge chooses **honest nulls** for densify by default.

Rules (hard)

Rule	Implementation
Never invent 0	<code>mag = magNd ? null : e.magnitude</code>
N/D is explicit	<code>magNd: true, magType → "N/D" if missing</code>
Hub-readable title	<code>M- - {place} when null</code>
Finite geometry required	<code>isSesPublishableEvent — lat/lon/depth/time/id</code>
Mag optional for densify	<code>null mag still published if hypocentre is finite</code>

language-81

```
export function hasFiniteMag(e: QuakeEvent): boolean {
  return e.magnitude !== null && Number.isFinite(e.magnitude);
}
// ...
const magNd = !hasFiniteMag(e);
// Never invent 0 for N/D — explicit null + M- title for hub rendering
const mag = magNd ? null : e.magnitude;
const title = magNd
  ? `M- - ${place || e.id}`
  : `M ${e.magnitude!.toFixed(1)} - ${place}`;
```

Two modes via `toSesEqCollection`

language-139

```
export function toSesEqCollection(
  events: QuakeEvent[],
  nodeId: FocusNodeId,
  opts: SesCollectionOpts = {},
): SesEqCollection {
  const publishable = events.filter(isSesPublishableEvent);
  const selected = opts.requireMag
    ? publishable.filter(hasFiniteMag)
    : publishable;
  // ...
  magNullPolicy: opts.requireMag
    ? "omit-unmaged"
    : "keep-null-never-zero",
}
```

Mode	Query	Policy string	Features
Default densify	(no flag)	keep-null-never-zero	All finite-hypocentre events; null mag kept
Stats-safe	? requireMag=1	omit-unmaged	Only finite mag

Metadata always reports:

- magNullCount — unmaged in publishable set (0 when requireMag)
- magFiniteCount — finite mags in publishable set
- requireMag — boolean
- count — **selected** feature count (not raw catalog count)

Feature shape for hub

JSON

```
{
  "id": "gossip-54411",
  "properties": {
    "mag": null,
    "magNd": true,
    "magType": "N/D",
    "title": "M— - Campi Flegrei",
    "net": "ov",
    "sources": "INGV-OV"
  },
  "geometry": { "type": "Point", "coordinates": [14.15, 40.83, 1.2] }
}
```

Hub contract notes (also in feed `metadata.note`):

- Min-mag filters must **skip** `mag == null` (`null ≠ 0`)
- UI should show **M—**, not M0
- Prefer default mode for swarm densify completeness

Wire points

Surface	Behavior
Dev Vite middleware	Parses <code>requireMag</code> , passes opts
Nitro <code>catalog.get.ts</code>	Same for production
TanStack <code>/api/ses/catalog</code>	Same
<code>mergeSesWithAuthorityNode</code>	Always keep-null-never-zero (no <code>requireMag</code> yet)
Schema docs	<code>docs/SES-HUB-SCHEMA.md</code> + <code>ses-catalog.schema.json</code>

ETag includes `rm` vs `all` so `requireMag` doesn't share cache with densify.

Verified live behavior (prior check)

Call	n	null mags	zeros
<code>window=7d default</code>	37	14	0
<code>requireMag=1</code>	23	0	0

What SES should do

1. **Densify / map** — use default feed; plot null-mag points; label M—.
2. **Min-mag / max-M / energy** — ignore or separate nulls; never coerce to 0.
3. **Optional** — if hub only wants finite-mag stats, call with `requireMag=1` (thinner CF swarm).
4. Read `metadata.magNullPolicy` + counts for UI honesty.

Bottom line: Policy is **option 3 from the SES smoke note** (keep explicit null) with an opt-out omit path. Densify stays complete under GOSSIP N/D load without lying about energy.